



BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

MediSync+ is an AI-powered mobile application designed to detect risky symptoms timely after discharge to prevent readmission. MediSync+ functions as a post-discharge risk detection that anticipates by using chatbot, asking after taking medicine and checking condition records through an orchestrator based multi-agent AI architecture. The total addressable market for digital post-discharge care in Southeast Asia is estimated at RM 7.2 billion, driven by ageing demographics, increasing medical conditions including but not limited to chronic and cancer disease burden, as well as rapid smartphone adoption. MediSync+ generates revenue through hospital and clinic SaaS (Software as a Service) subscriptions and data analytics licensing. With an initial focus on Malaysian public and private hospitals, the platform targets RM 15.7 million in revenue by Year 3.

2. Product Overview / Solution

Key Features:

- **Medicine Management:** Automated scheduling, push notifications for reminders and real-time adherence scoring.
- **Follow-up Coordination:** Integration with AppointmentDao to track upcoming visits and push notifications for reminders.
- **Today's Checklist:** Daily tasks generated by DailyChecklistGeneratorWorker, with ChecklistAgent handles vital recording tasks (e.g.: weight, blood pressure).
- **Interactive Chatbot:** ChatAgent powered by Z AI's GLM using tools to access patient's historical databases, provide clinical advices, record patient's feedbacks (e.g.: vital symptoms).
- **Condition Follow-up:** For all features above, emergency SosDialog will be triggered immediately once critical alerts or risky symptoms are detected by agents.



- **Symptoms detection:**
 - **Chatbot:** Tool that can be accessed by the user to ask questions whenever the user feels uncomfortable
 - **After medicine checkup:** A daily checkup that ensures the user's health is that it's most optimum condition
 - **Condition record:** Anticipating the risk of the user's health based on the records that are inside the system

Technology Stack:

- Multi-agent framework built on Llama-3.3-70B-versatile and optimized via Groq for scalable, efficient intelligence.
- LangSmith for real-time LLM performance monitoring and evaluation of LLM performance
- A worker-driven automation framework handles background processes such as critical medication and appointment reminders
- Modern Android Stack: Jetpack Compose (UI), Hilt (Dependency Injection), and Room (Secure Local Persistence).

3. Market Research & Market Dynamics

3.1 Problem Landscape

Despite advances in in-hospital care, many patients experience fragmented workflows, untracked symptoms and a lack of continuous physiological monitoring, meaning subtle symptoms and early signs of deterioration often go undetected until a medical crisis occurs. Without a structured safety net, the absence of real-time data leaves clinicians unable to provide proactive oversight, transforming a period meant for healing into a high-risk window for rapid health regression. At the same time, medical teams often lack real-time visibility into patient health after discharge and between appointments, often catching complications only when they become emergencies.

3.2 Market Niche

- Patients side: MediSync+ targets discharged patients and their caregivers, specifically those elderly, chronic disease patients and high-risk disease patients such as cancer patients who are having high-risk conditions or requires continuous structured care.
- Healthcare provider side: MediSync+ targets private hospitals, specialist clinics, government hospitals departments (e.g.: cardiology, cancer, etc.) seeking to reduce readmission rates and improve post-discharge care coordination.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	Digital post-discharge & remote patient monitoring market, Southeast Asia	RM 7.2 billion
SAM	Smartphone-owning post-discharge patients in Malaysia + Singapore	RM 980 million
SOM	Realistic capture via hospital partnerships	RM 15.7 million

3.4 Market Dynamics & Placement:

There were few key dynamics that shape MediSync+ entry to this market.

- **Aging Population:** The rapidly growing elderly population with 56.2 million elderly citizens aged 65+ in Southeast Asia (2023) and elderly citizens aged 60+ comprising 12% of Malaysia's population (April 2026), the market faces a critical surge in demand for post-discharge monitoring and long-term care support.
- **Rising Chronic Disease Burden:** The prevalence of chronic conditions such as diabetes, cardiovascular diseases, and hypertension is steadily increasing, creating a need for structured and ongoing care management beyond hospital settings.
- **Growth in High-Risk Patient Group:** Increasing numbers of cancer patients and individuals undergoing intensive treatments require closer follow-up, symptom tracking, and coordinated care after discharge.
- **AI Acceptance:** Growing trust in AI-assisted triage to reduce the burden on medical staff.
- **Smartphone Penetration:** Malaysia's smartphone penetration up to 99%, with strong Android adoption — directly aligned with MediSync+'s Android-first deployment strategy.
- **Healthcare Digitalisation:** KKM's MyHEALTH and MERS999 initiatives signal strong government appetite for digital health infrastructure, opening procurement channels for compliant health tech platforms.

4. Value Proposition & Competitive Advantage

MediSync redefines post-discharge medical care by moving from reactive to proactive management by turning the patient's smartphone into a continuous clinical monitoring station. Unlike generic wellness apps or simple medication reminder tools, MediSync+ deploys a coordinated multi-agent AI system — seven specialist agents working in parallel — that replicates the cognitive workflow of a clinical team at a fraction of the cost.

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	High	● The Race for Market Share: Rivals are aggressively targeting high-density hubs like Kuala Lumpur to capture the RM 84 million SOM through rapid feature expansion. ● Specialized AI Entrance: New players like Hippocratic AI are focusing on empathetic LLM interactions, forcing a race for clinical accuracy and integrated recovery workflows.
Threat of New Entrants	Medium	● Technical & Integration Moat: Establishing a direct sync with hospital Historical Ward and Follow-up Databases creates a high barrier that simple "app-only" startups cannot easily cross. ● Agentic Complexity: Building a multi-agent system capable of "reasoning" through unstructured discharge notes requires specialized clinical logic, which is harder to replicate than standard notification features.
Bargaining Power of Buyers	Medium	● Institutional Dependency: Hospitals hold significant power as they provide the foundational clinical data and summaries required for the system's reasoning baseline.

		● High Switching Costs: Once MediSync+ is integrated into a hospital's digital care workflow, the logistical effort to replace it with a new system becomes a major deterrent for buyers.
Bargaining Power of Suppliers	Medium	● Data Monopoly: Hospitals act as the primary "suppliers" of the raw medical data needed to run the specialized sub-agents. ● Cloud Infrastructure Dependence: The system relies on specific LLM and cloud providers to maintain the processing power needed for real-time agentic orchestrations.
Threat of Substitutes	High	● Entrenched Manual Methods: The most common substitute remains the status quo—fragmented paper discharge summaries and manual home care instructions. ● Labor-Intensive Services: High-touch human caregiving services (like Homage) act as a substitute by providing physical observation instead of a digital "virtual nurse".

Competitor Comparison Table:

Competitor	Strength	Weakness	MediSync+'s Advantage
Hippocratic AI	Provides high-quality, empathetic patient education through advanced voice and chat interfaces. Focuses on safe, patient-facing dialogue using Large Language Models (LLMs) specifically trained for healthcare.	Lacks specialized multi-agent orchestrator to manage structured clinical workflows like medication schedules. It operates as a conversation tool and is not directly synced with the patient's hospital follow-up database.	Our system uses an Orchestrator Agent to manage distinct sub-agents, ensuring that fragmented discharge notes are converted into executable medical plans. We maintain a closed-loop system that updates the hospital's database in real-time, which a chat-only AI cannot do.
Biofourmis	Delivers clinical-grade Remote Patient Monitoring (RPM) by utilizing specialized wearable sensors for vitals tracking. Strong focus on hospital-at-home models for complex chronic condition management.	Requires expensive hardware and clinical infrastructure, making it difficult to scale to low-income populations. Highly dependent on sensors; if a patient forgets to wear the device, the system's data flow stops.	MediSync+ is a software-only, highly scalable solution that provides predictive care through agentic reasoning on any smartphone. We specifically target SDG 10 by providing affordable care to those who cannot afford expensive monitoring hardware.

Medisafe	A high-functioning digital pillbox with reliable, rule-based notifications for medication adherence. Features a large, global user base and a simple, user-friendly interface for manual tracking.	Relies entirely on manual data entry by the patient, leading to "reminder fatigue" and potential entry errors. It is an isolated app that does not alert hospital staff or doctors when a patient's health trajectory begins to decline.	Our Risk Trajectory Agent automatically predicts health deterioration by analyzing checklist data, rather than just waiting for a missed pill alert. We eliminate manual entry by automating the ingestion of unstructured discharge summaries directly from the ward.
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6. Business Model Canvas

6.1. Customer Segmentation

1. **Post-Discharge Patients:** Discharged patients who are having high-risk conditions or require continuous monitoring.
2. **Healthcare Providers (Hospitals & Clinics):** Private hospitals, specialist clinics, government hospitals departments (e.g.: cardiology, cancer, etc.) seeking to reduce readmission rates and improve post-discharge care coordination.
3. **Caregivers:** Caregivers who want visibility and guide into patient recovery workflow and to receive alerts when clinical risk escalates.

6.2. Revenue Stream

Primary Revenue — Hospital SaaS Subscription



- **Per-ward/per-department licensing:** RM 2,500–6,000/month per hospital unit depending on patient volume.
- **Enterprise hospital group licensing:** RM 30,000–80,000/month for group-wide access.

Secondary Revenue

- **Data Analytics & Insights Licensing:** Anonymised, aggregated population health insights licensed to insurance companies and pharmaceutical firms — RM 10,000–50,000/report.
- **Professional Services:** Hospital onboarding, integration with existing HIS/EMR (Hospital Information System / Electronic Medical Record) systems, clinical staff training — project-based pricing.

Recurring Maintenance & Support (The "Reliability" Stream)

- **Annual Maintenance for Integrations:** 15% – 20% of the initial integration fee, billed annually.
- **Premium Support:** RM 1,500 – RM 5,000/month per hospital site.

6.3 Cost Structure

Cost Drivers	Cost Type
Platform development (Kotlin/Android), AI agent engineering, ongoing R&D	Fixed Cost
LLM API inference costs (Groq llama-3.3-70b-versatile), cloud hosting, data infrastructure	Variable / Usage-based Cost

Hospital sales cycle support, clinical validation studies, regulatory compliance (MDA)	Sales & Compliance Overhead
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6.4 Key Partners

Internal	External
Groq (LLM API provider)	Malaysian public and private hospital networks
LangSmith (LLM performance monitoring)	Cardiology & chronic disease specialist associations (e.g., National Heart Association of Malaysia)
Hospital Information System (HIS) vendors for EMR integration	Medical Device Authority (MDA) Malaysia — Regulatory body
	Insurance providers (for reimbursement pathway development)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Regulatory / MDA classification delay	High	Medium	AI positioned as decision-support (not autonomous diagnosis); clinician review gate on all EMR proposals; early MDA pre-submission engagement.
AI hallucination / clinical error	High	Low	ResponseValidator layer validates all LLM outputs; full audit trail; PHI never sent to LLM; human-in-the-loop for all clinical decisions.

Patient digital literacy barrier	Medium	Medium	Caregiver proxy access; hospital nurse-led onboarding session; BM language support in roadmap.
LLM API cost scaling	Medium	Medium	Rate-limiting agent calls, caching common responses, local model fallback for non-critical queries.

7. Implementation & Financials

7.1 Go-to-Market Strategy

MediSync+ adopts a hospital-partnership-led rollout strategy. The go-to-market is driven by clinical outcome evidence rather than consumer marketing. Phase 1 targets two private hospital pilot partners, with a structured 90-day readmission reduction and patients & medical staff post-discharge service satisfaction study. Outcomes data becomes the primary sales tool for subsequent hospital and public sector outreach.

Patient acquisition is therefore hospital-mediated: discharge coordinators and ward nurses introduce MediSync+ to eligible patients at the point of discharge, removing the need for consumer marketing spend. This dramatically reduces customer acquisition cost compared to a direct-to-consumer model.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Pilot Launch	2 private hospital partners, ward pilot; IRB ethics approval	Month 1–6	150 active patients; baseline readmission data and feedback from

			discharged patients and medical staff collected
Expansion — Private	Expand to 5 private hospitals nationally; add hypertension & post-surgical modules	Month 6–12	750 active patients; 3 hospital SaaS contracts signed
Public Sector Tender	Submit MOH pilot tender with readmission reduction, patients and medical staff feedbacks evidence; BM localisation	Month 12–18	First government hospital contract; 2,500 patients
Regional Expansion	Singapore & Thailand private hospital market entry; multi-language support	Month 18–30	5,000+ patients; break-even achieved
TOTAL INVESTMENT	RM 8.4M across 30 months	30 Months	Market leadership in SEA post-discharge AI care

7.3 Revenue Projections (e.g 3-Year)

<i>Metric</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>
Hospital Partners	2	8	20
Active Patients	300	2,500	8,000
Hospital SaaS Revenue	RM 0.6M	RM 4.8M	RM 12.0M
Analytics Licensing	RM 0.0M	RM 0.3M	RM 1.2M
Professional Services	RM 0.2M	RM 0.5M	RM 2.5M
Total Revenue	RM 0.8M	RM 5.6M	RM 15.7M

Net Position	RM (3.9M)	RM (1.3M)	RM +2.9M
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7.4 Key Financial Milestones

- Month 6: First pilot readmission reduction data published; 150 active CHF patients enrolled across 2 private hospitals.
- Month 12: 3 signed SaaS contracts; expansion to 750 active patients across 8 hospital units nationally.
- Month 18: First public hospital MOH pilot contract; 2,500 active patients; 30-day readmission rate reduction of $\geq 20\%$ and patients & medical staff positive feedback of $\geq 95\%$ demonstrated across pilot hospitals.
- Month 30: Cash-flow positive; regional expansion to Singapore underway; 5,000+ active patients; feasibility review for Series A funding.